
PRINCIPLES TO FINANCE



*The Finance of the Firm:
An Introduction for First-Year Students*

A TEXTBOOK FOR FIRST-YEAR STUDENTS

BIJOY?!



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An Introduction for First-Year Students*

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TO THE READER

This book teaches how a business finds, uses and controls its money. It assumes no earlier knowledge of business. If you can do arithmetic and basic algebra, you can follow every step. A word you have not met will be explained when it first appears.

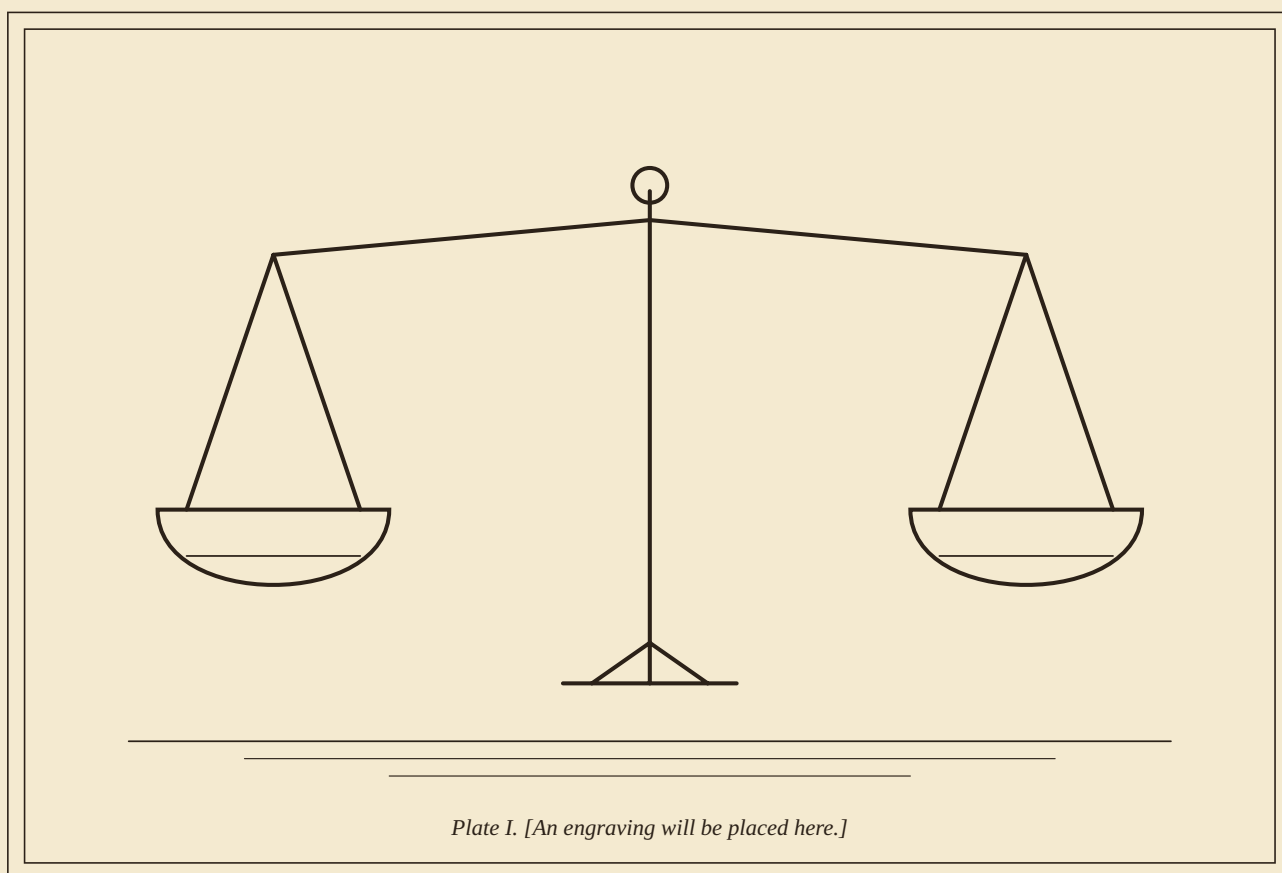
Each page has a wide column of short notes on the left. An italic note repeats the idea of the paragraph beside it in one line. A note in upright type, with a bold word at its start, gives the meaning of a new term. A word in bold in the main text is a new term, set in bold once only. Small raised numbers point to footnotes at the foot of the page. All terms are collected again in the glossary at the back.

Money is written in Bangladeshi taka, shown as Tk. Large sums use the South Asian grouping: Tk. 1,00,000 means one lakh, which is one hundred thousand, and Tk. 10,00,000 means ten lakh, which is one million. Laws, taxes and rates change; where a figure is an assumption it is said to be one. The Bakery in the case files is an invented business, used for illustration. It is not a real firm.

Each chapter begins with a puzzle, builds an idea in steps, and ends with a summary, a problem, common mistakes, a true story from history, and review questions. Short answers are printed at the back. Work each calculation with a pencil before you read the answer.

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CHAPTER I

The Purpose of Finance

Money is easy to count and hard to use well.



This chapter explains what finance is, what a financial manager decides, why the aim of a firm is wealth and not only profit, and why owners and managers may want different things.

A business can earn a profit and still run short of money.

Finance is about finding money and using it well.

Finance. The way a business finds, uses and controls its money.

Accounting looks back; finance looks forward.

Financial manager. The person who decides how a firm finds and uses its money.

Four kinds of decision cover the whole task.

Investment decision. A choice about which assets, such as machines or shops, the firm should buy.

Financing decision. A choice about how to raise money, from owners or from lenders.

Dividend. A share of profit paid out to the owners of a company.

A PUZZLE ABOUT MONEY

A baker sells bread for a month. At the end, her records show a profit of Tk. 20,000. On Friday the flour supplier asks to be paid, and the bank balance is Tk. 4,000. How can a business make a profit and still be unable to pay its bills? The answer lies in the way money moves through a business. The study of that movement is the subject of this book.

WHAT FINANCE MEANS

Finance means the way a business finds money, uses money and controls money. A business needs money to begin, for an oven, flour and rent. It needs more money to continue, for wages and repairs. Finance asks three plain questions. Where will the money come from? Where will it be placed? How will we know it was placed well?

Finance is close to two other subjects. Economics explains how people and markets make choices when resources are limited. Accounting records what has happened in a business. Finance uses both: it takes the records of the past and the ideas of economics to decide what the business should do next.

The person who makes these choices is the **financial manager**. In a small firm the owner does this work. In a large firm a team does it. The financial manager is also a manager in the ordinary sense, so finance and management cannot be separated: every plan of the firm, whether to hire, to expand or to buy, needs money, and the financial manager says whether the money can be found and what it will cost.¹

Before moving on. Finance means finding, using and controlling money. It looks forward, where accounting looks back. The financial manager makes the choices about money.

THE DECISIONS OF THE FINANCIAL MANAGER

The first decision concerns what to buy. An **investment decision** means a choice about which assets the firm should buy, such as a machine, a shop or a delivery van. An asset is anything of value that the firm owns. The second decision concerns where the money comes from. A **financing decision** means a choice about how to raise money: from the owners, from lenders, or from the profits the firm has kept.

The third decision concerns daily money. The working-capital decision means a choice about how much cash, stock and credit the firm needs from day to day to keep running. The fourth is the dividend decision. A **dividend** is a share of profit paid to the owners. The firm must decide how much profit to pay out and how much to keep for growth.

THE FOUR DECISIONS

Decision	The question it answers	Example
Investment	What should we buy?	A second oven
Financing	Where does the money come from?	A bank loan or owners' savings
Working capital	How do we keep daily money flowing?	How much flour to hold in stock
Dividend	How much profit do we pay out?	Pay owners now or keep the money for growth

Good finance keeps the business running and growing.

The firm needs one clear aim to judge its decisions.

Profit maximisation. Aiming for the largest possible profit.

Go slowly here.

Wealth is the value of the owners' investment.

Wealth maximisation. Aiming to make the owners' investment as valuable as possible.

When owners hire managers, their interests may differ.

Shareholders. The people who own shares, and so own part of a company.

Principal. A person who asks another person to act for them.

Agent. A person who acts on behalf of another.

Agency problem. The risk that agents act for themselves instead of for the owners.

Agency costs. The costs of watching agents and of the losses when

Read each row from left to right: the decision, the question behind it, and a case from a bakery. Later chapters teach how to make each one well.

Finance affects how well a business runs and whether it survives. A firm with the right money at the right time can buy materials, pay workers and keep machines busy. A firm that lacks money may stop production even when customers want its goods. A firm that borrows too much may be unable to pay its lenders. A firm that keeps idle cash earns nothing from it. In each case the quality of financial decisions shows in the firm's efficiency and in its success.

THE GOAL OF THE FIRM

Decisions need a standard. Without one, we cannot say whether a choice was good. The oldest standard is **profit maximisation**, which means aiming for the largest profit. It sounds sensible, but it has three weaknesses.

Picture two market stalls. Stall A will earn Tk. 100 profit this year and Tk. 100 next year. Stall B will earn Tk. 50 this year and Tk. 200 next year. Which has the larger profit? Stall B, in total. Yet the owner who needs money now may prefer A. The first weakness is that profit maximisation ignores *when* the money arrives. Chapter II explains why money received earlier is worth more.

The second weakness is risk. A profit that is almost certain and a profit that depends on luck count as equal, though they are not. The third is that profit, as accountants measure it, is not the same as cash, as the puzzle showed. A firm can also raise this year's profit by cutting maintenance or training, and so harm the years after.

A better standard is **wealth maximisation**. It means aiming to make the owners' investment as valuable as possible. The value of a firm depends on the cash it will give to its owners in future, the timing of that cash, and the risk of receiving it. This one idea includes the three things that profit leaves out. For this reason, profit is important but is not the ultimate goal.

You should now be able to say: *a firm should choose the plan that adds most to the value of the owners' investment, taking account of time and risk, and not simply the plan with the largest profit this year.*

Pause and check. Why might Stall B be worth more than Stall A, even though the owner who needs cash now prefers A?

OWNERS AND MANAGERS

In a large company the owners cannot run the firm. The owners are called **shareholders**. They hire managers to run it for them. In finance the owner is called the **principal** and the manager is called the **agent**. The principal wants the value of the firm to grow. The agent may want a high salary, a comfortable life or a larger firm, even when these do not raise the owners' wealth.

This gap is the **agency problem**. It has a cost. The owners may spend money to watch the managers, for example by hiring auditors, and some bad decisions may still be made. These costs together are called **agency costs**. Owners reduce the problem in two main ways: by linking the manager's pay to the firm's value, for instance through bonuses paid in shares, and by monitoring, through audits, reports and a board of directors.

Exceptions matter. In a business run by its single owner there is no agency problem, since principal and agent are the same person. The problem also exists

they act against the owners.

between lenders and owners, because owners may take risks with money that lenders supplied.



WORKED CASE: THE BAKERY

WORKED EXAMPLE: WHY DID CASH FALL WHEN THE BAKERY MADE A PROFIT?

SITUATION. *Case file, illustrative.* In January 2025 the owners put Tk. 5,00,000 into the Bakery. Oven and fittings cost Tk. 3,50,000, leaving cash of Tk. 1,50,000. In its first month the Bakery sold goods worth Tk. 1,50,000, of which Tk. 60,000 was sold to a café on 30 days' credit. Costs of Tk. 1,30,000 were paid in cash.

QUESTION. What was the profit, and what happened to cash?

REASONING. Profit is sales minus costs: $1,50,000 - 1,30,000 = \text{Tk. } 20,000$. Cash received was $1,50,000 - 60,000 = \text{Tk. } 90,000$. Cash paid was Tk. 1,30,000. Cash change is $90,000 - 1,30,000 = \text{minus Tk. } 40,000$.

CONCLUSION. The Bakery made a profit of Tk. 20,000, yet its cash fell by Tk. 40,000, from 1,50,000 to Tk. 1,10,000. The Tk. 60,000 owed by the café is profit that has not yet become cash.

CHECK. Profit 20,000 less the unpaid 60,000 gives minus 40,000, which matches the cash change.

CLOSING THE CHAPTER

SUMMARY

Finance is the way a firm finds, uses and controls money. The financial manager makes investment, financing, working-capital and dividend decisions. The sound goal is wealth maximisation, because it counts time and risk. Where owners and managers are different people, agency costs arise and must be controlled.

THINK IT THROUGH

A manager can raise this year's profit by Tk. 50,000 by stopping all repairs to the ovens. Explain why this may reduce the owners' wealth.

WHERE PEOPLE GO WRONG

Many students think profit and cash are the same; the case above shows they are not. Others think wealth maximisation means ignoring profit; it does not, since profit is one source of future cash.

IN HISTORY

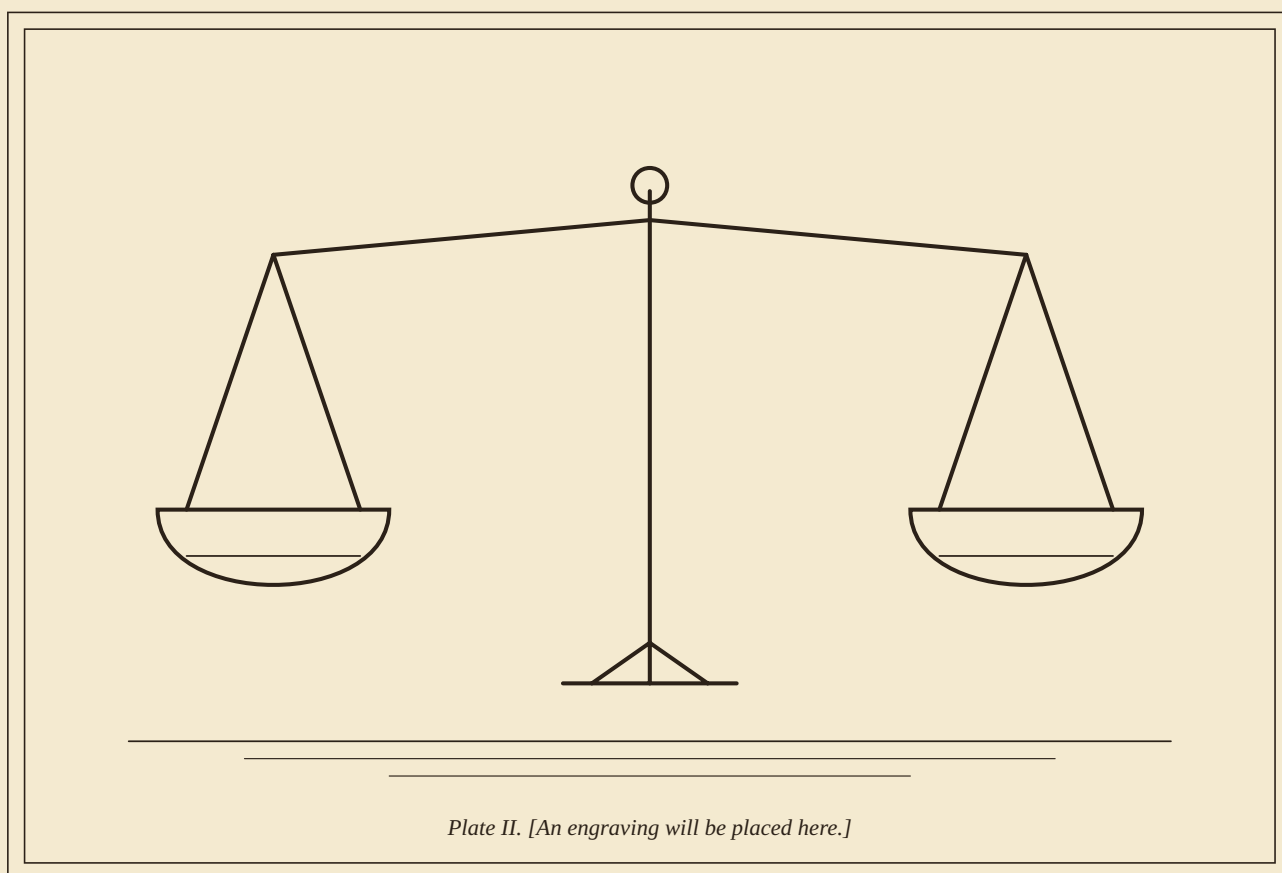
In 1602 the Dutch East India Company offered shares to the public, and ordinary people became part-owners of a business run by others far away. In 1776 Adam Smith wrote that directors of such companies, handling other people's money, could hardly be expected to watch it as anxiously as owners watch their own.²

REVIEW QUESTIONS

1. What is meant by finance?
2. Name the four decisions of a financial manager and say what each answers.
3. Why is profit maximisation not the ultimate goal of a firm?

Profit and cash are different things.

¹ Management is the wider task of organising people and work to reach a goal. Finance is the part of that task that deals with money.



CHAPTER II

Money Across Time

A sum in your hand is worth more than the same sum promised later.



This chapter explains why money has a time value, and how to move a sum forward to its future value or back to its present value, for a single sum, for equal payments, and for payments that differ.

² Smith, *The Wealth of Nations*, Book V. The point was formalised by Jensen and Meckling in 1976.

Would you take Tk. 1,000 today or Tk. 1,000 in a year?

Money today can earn; money later cannot.

Time value of money. The idea that a sum of money is worth more now than the same sum later.

Go slowly here.

Interest. The payment a borrower makes for the use of money, shown as a percentage of the sum.

Compound interest. Interest calculated on the original sum plus the interest already earned.

Future value pushes money forward; present value pulls it back.

Future value. What a sum of money today will be worth at a later date, after earning interest.

Present value. What a sum due in the future is worth today.

Discount rate. The yearly rate used to reduce a future sum to its present value.

Common slips with single sums.

A PUZZLE ABOUT WAITING

A friend offers you a choice: Tk. 1,000 today, or Tk. 1,000 in one year. Most people take the money today. Now the offer changes: Tk. 1,000 today, or Tk. 1,100 in one year. The choice is harder. Somewhere between 1,000 and 1,100 lies a point where you are indifferent. That point is the key to this chapter.

THE TIME VALUE OF MONEY

The **time value of money** means that a sum of money is worth more now than the same sum later. There are three reasons. Money held today can be placed in a bank or a business and earn a return. Prices usually rise, so later money buys less. And a promise of future money may not be kept, so waiting carries risk.

Suppose you place Tk. 1,000 in a bank that pays 10% a year. **Interest** means the payment for the use of money. After one year you hold $1,000 + 100 = \text{Tk. } 1,100$. In the second year the bank pays 10% on 1,100, not on 1,000. You hold $1,100 + 110 = \text{Tk. } 1,210$. Interest earned on earlier interest is called **compound interest**.

You should now be able to say: *compound interest grows a sum faster than simple interest, because each year's interest earns interest too.*

MOVING A SUM FORWARD AND BACK

The **future value** (FV) is what a sum today will be worth later. The **present value** (PV) is what a future sum is worth today. The **discount rate** is the rate used to move a future sum back to today. We write r for the rate as a decimal and n for the number of years.

Rule: $\text{future value} = \text{present value} \times (1 + r)^n$. $\text{Present value} = \text{future value} \div (1 + r)^n$. The two are opposites.

WORKED EXAMPLE: SINGLE SUM

GIVEN. $\text{PV} = \text{Tk. } 1,000$; $r = 10\%$ a year; $n = 2$ years.

FIND. the future value, and then the present value of that future value.

FORMULA. $\text{FV} = \text{PV} \times (1 + r)^n$; $\text{PV} = \text{FV} / (1 + r)^n$.

STEPS. $\text{FV} = 1,000 \times 1.10^2 = 1,000 \times 1.21 = \text{Tk. } 1,210$. $\text{PV} = 1,210 / 1.21 = \text{Tk. } 1,000$.

ANSWER. Tk. 1,000 today equals Tk. 1,210 in two years at 10%.

CHECK. This matches the year-by-year result above: 1,100, then 1,210. Moving forward and back returns the starting sum.

WORKED EXAMPLE: A MESSIER CASE: TK. 25,000 FOR 5 YEARS AT 12%

GIVEN. $\text{PV} = \text{Tk. } 25,000$; $r = 12\%$; $n = 5$.

FIND. FV.

FORMULA. $\text{FV} = \text{PV} \times (1 + r)^n$.

STEPS. $1.12^5 = 1.7623$. $\text{FV} = 25,000 \times 1.7623 = \text{Tk. } 44,059$.

ANSWER. Tk. 25,000 grows to about Tk. 44,059 in five years.

CHECK. Simple interest would give $25,000 + 5 \times 3,000 = \text{Tk. } 40,000$. Compound interest must be larger, and it is.

Common slips: using 10 instead of 0.10 in the formula; forgetting that n counts years, so 6 months is $n = 0.5$; and dividing when the question asks you to multiply.

Before moving on. Multiply by $(1 + r)^n$ to go forward; divide to go back. Compound interest earns interest on interest.

EQUAL PAYMENTS: ANNUITIES

An annuity is a series of equal payments at equal gaps.

Annuity. A series of equal payments made at equal intervals.

Ordinary annuity. An annuity whose payments are made at the end of each period.

Annuity due. An annuity whose payments are made at the start of each period.

An **annuity** is a series of equal payments made at equal intervals. Rent, loan instalments and regular savings are annuities. In an **ordinary annuity** each payment falls at the end of its period. In an **annuity due** each payment falls at the start. A third kind, the perpetuity, never ends; its present value is the payment divided by the rate.

The formulas are: FV of an ordinary annuity = payment $\times [(1 + r)^n - 1] / r$. PV of an ordinary annuity = payment $\times [1 - (1 + r)^{-n}] / r$. For an annuity due, multiply the ordinary result by $(1 + r)$, because every payment comes one period earlier and so earns or loses one more period of interest.

WORKED EXAMPLE: SAVING Tk. 1,000 A YEAR FOR 3 YEARS AT 10%

GIVEN. payment Tk. 1,000 at the end of each year; $r = 10\%$; $n = 3$.

FIND. the future value, the present value, and the future value if payments start at the beginning of each year.

FORMULA. $FV = P \times [(1 + r)^n - 1] / r$; $PV = P \times [1 - (1 + r)^{-n}] / r$.

STEPS. Factor for FV = $(1.331 - 1) / 0.10 = 3.31$, so FV = Tk. 3,310. Factor for PV = 2.48685, so PV = Tk. 2,486.85. Annuity due: $3,310 \times 1.10 = \text{Tk. } 3,641$.

ANSWER. The ordinary annuity grows to Tk. 3,310. Its present value is about Tk. 2,487. The annuity due grows to Tk. 3,641.

CHECK. Add each payment by hand: $1,000 \times 1.21 + 1,000 \times 1.10 + 1,000 = 1,210 + 1,100 + 1,000 = \text{Tk. } 3,310$.

PAYMENTS THAT DIFFER: MIXED STREAMS

A mixed stream is handled one payment at a time.

Mixed stream. A series of payments that are not all equal.

A **mixed stream** is a series of payments that are not all equal. No short formula exists. We move each payment separately and add the results. For the present value we divide each payment by $(1 + r)^n$ for its own year. For the future value we multiply each by $(1 + r)$ raised to the years remaining.

WORKED EXAMPLE: PRESENT VALUE OF A MIXED STREAM

GIVEN. Year 1 Tk. 1,000; Year 2 Tk. 2,000; Year 3 Tk. 3,000; discount rate 10%.

FIND. the present value.

FORMULA. $PV = \text{sum of each payment} / (1 + r)^n$.

STEPS. Year 1: $1,000 \times 0.9091 = 909.09$. Year 2: $2,000 \times 0.8264 = 1,652.89$. Year 3: $3,000 \times 0.7513 = 2,253.94$.

ANSWER. The total present value is Tk. 4,815.93.

CHECK. The total is below the plain sum of 6,000, as it must be, because later money is worth less today.

Exceptions deserve a note. If interest is added more often than once a year, say every month, use r divided by the number of periods and n multiplied by it. When comparing two streams, discount both at the same rate; the stream with the higher present value is worth more today.

Case file, illustrative. The Bakery sets aside Tk. 1,00,000 at the end of each of three years, in an account paying 10%. The ordinary annuity gives $1,00,000 \times 3.31 = \text{Tk. } 3,31,000$, enough toward a second oven.

The case file: saving for a new oven.

CLOSING THE CHAPTER

SUMMARY

Money has a time value because it can earn, prices rise and promises carry risk. Future value moves money forward and present value moves it back. Annuities use short formulas; mixed streams are moved one payment at a time.

THINK IT THROUGH

Stream P pays Tk. 5,000 at the end of each of 3 years. Stream Q pays 2,000, 4,000 and 7,000 in those years. Find the present value of each at 10% and say which is worth more today.

WHERE PEOPLE GO WRONG

Students add payments from different years as if they were equal. Others use an ordinary annuity formula for payments made at the start of the period.

IN HISTORY

In 1202 the Italian mathematician Leonardo of Pisa, called Fibonacci, published *Liber Abaci*, which taught merchants of Europe the Hindu-Arabic numerals. It included problems on interest and on the value of money at different times, tools for the traders of his day.

REVIEW QUESTIONS

1. What is meant by the time value of money?
2. What is an annuity, and what are its kinds?
3. Find the present values of streams P and Q at 10%. Which is better?
4. Tk. 1,00,000 is saved at the end of each year for 5 years at 8%. What is the future value?

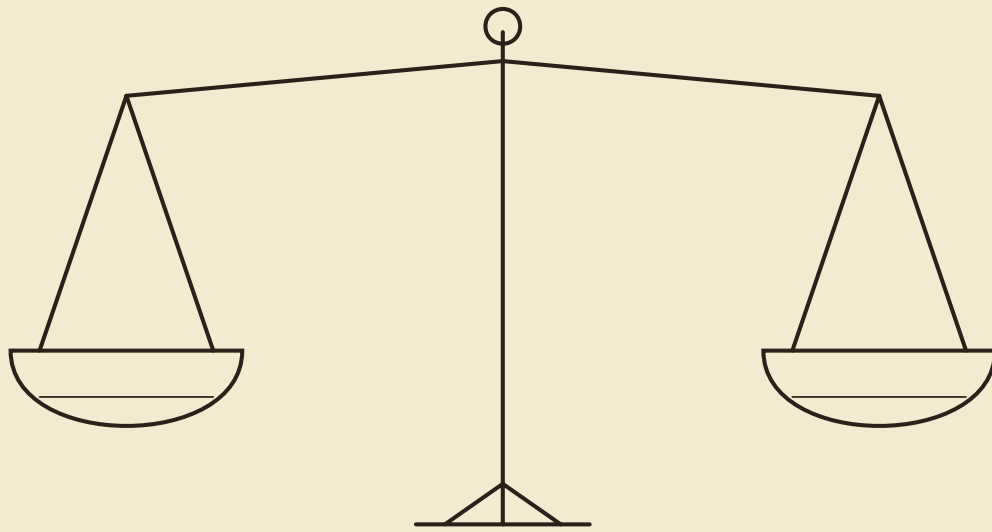


Plate III. [An engraving will be placed here.]

CHAPTER III

Judging Projects

A machine costs money now and pays back slowly; is it worth buying?



This chapter explains capital budgeting: why it matters, the steps of the process, which cash flows count, and how to judge projects by payback and by net present value.

Two machines cost the same but pay back differently.

Large, long-lasting spending needs careful judgement.

Capital budgeting. Planning and judging large spending whose benefits last for several years.

Capital budgeting follows six steps in order.

Only extra cash caused by the project counts.

Cash flow. Money actually received or paid by the firm.

Incremental. Extra; the change in cash caused by choosing the project.

Sunk cost. Money already spent that cannot be recovered whatever is decided.

Payback asks how soon the money returns.

Payback period. The time a project takes to return its initial cost in cash.

NPV measures the wealth a project adds.

Net present value. The present value of a project's cash inflows minus its initial cost.

A PUZZLE ABOUT TWO MACHINES

Machine A and Machine B each cost Tk. 10,000. A returns Tk. 5,000 a year for three years. B returns Tk. 2,000, then 4,000, then 10,000. Both return Tk. 15,000 and Tk. 16,000 in total. Which should the owner buy? A simple count of total money received favours B, but A returns money sooner. We need a method that treats both the size and the timing of money fairly.

WHAT CAPITAL BUDGETING IS

Capital budgeting means planning and judging large spending whose benefits last for several years, such as a new oven, a shop or a factory. It is necessary for four reasons. The sums are large. The benefits last a long time. Such decisions are hard to reverse, since a machine cannot be returned once installed. And they shape what the firm can do in future, so a poor choice weakens the firm for years.

THE STEPS OF THE PROCESS

First, the firm generates proposals, with ideas coming from managers, workers and customers. Second, it estimates the cash flows each proposal will cause. Third, it evaluates each proposal using a technique. Fourth, it selects and approves the best. Fifth, it carries out the project. Sixth, it reviews the results later and compares them with the estimates, so that future estimates improve.

CASH FLOWS THAT COUNT

A **cash flow** is money actually received or paid. The project's cash flows come in three groups: the initial outlay, which is the cost of starting; the operating cash inflows, which are the extra yearly receipts after tax; and the terminal cash flow, such as the sale of the machine at the end.

Only **incremental** cash flows count, which means the change caused by the project. A **sunk cost** is money already spent, such as a survey paid for last year; it is ignored, since it cannot be recovered. A use of an owned resource counts at what it could earn elsewhere. Depreciation is not a cash payment, so it is not counted as a cash flow, though it affects the tax paid.

TWO TECHNIQUES

The **payback period** (PBP) is the time a project takes to return its initial cost. The rule is to accept a project if its payback is shorter than a limit the firm has set, and to prefer the shorter payback. Its strength is simplicity and its attention to how soon the money returns. Its weakness is serious: it ignores the time value of money and it ignores all cash after the payback date.

The **net present value** (NPV) is the present value of the cash inflows minus the initial cost. The rule is to accept a project if NPV is greater than zero, and to prefer the project with the highest NPV. A positive NPV means the project earns more than the discount rate, and so adds to the owners' wealth. This links back to Chapter I: NPV is wealth maximisation made measurable.

Other techniques exist. The internal rate of return finds the discount rate at which NPV is zero. The profitability index divides the present value of inflows by the cost. They are met in later courses.

WORKED EXAMPLE: MACHINES A AND B AT A DISCOUNT RATE OF 10%

GIVEN. cost Tk. 10,000 each. A: 5,000, 5,000, 5,000. B: 2,000, 4,000, 10,000. Rate 10%.

FIND. payback period and NPV of each; which to prefer.

FORMULA. Payback = full years + (cost still unpaid / cash flow of the next year).

NPV = sum of PV of inflows - initial cost.

STEPS. A pays back in $10,000 / 5,000 = 2.0$ years. B returns 2,000 and 6,000 by the end of years 1 and 2, so 4,000 is unpaid; $4,000 / 10,000 = 0.4$, so payback is 2.4 years. NPV of A = $5,000 \times 2.48685 - 10,000 = 12,434.26 - 10,000 = \text{Tk. } 2,434.26$. NPV of B = $2,000 \times 0.9091 + 4,000 \times 0.8264 + 10,000 \times 0.7513 - 10,000 = 12,637.11 - 10,000 = \text{Tk. } 2,637.11$.

ANSWER. A has the shorter payback (2.0 against 2.4 years), but B has the higher NPV. The two methods disagree. NPV counts timing and all cash flows, so B is preferred as the better addition to wealth.

CHECK. B's large final payment of 10,000 is worth Tk. 7,513.15 today, which more than makes up for its slower start.

WORKED EXAMPLE: THE BAKERY'S SECOND OVEN AT 12%

GIVEN. cost Tk. 5,00,000. Inflows: Year 1 1,80,000; Year 2 2,00,000; Year 3 2,20,000; Year 4 1,50,000. Rate 12%.

FIND. NPV and the decision.

FORMULA. NPV = sum of inflow / $(1 + r)^n$ - cost.

STEPS. $1,80,000 \times 0.8929 = 1,60,714$; $2,00,000 \times 0.7972 = 1,59,439$; $2,20,000 \times 0.7118 = 1,56,592$; $1,50,000 \times 0.6355 = 95,328$. Total = 5,72,072. NPV = total - 5,00,000.

ANSWER. NPV is about Tk. 72,072. It is positive, so the project is acceptable.

CHECK. The plain sum of inflows is 7,50,000, and the discounted total is lower, as expected.

A second case, with messier numbers.

Exceptions and slips. When two projects exclude each other, as when only one oven fits the shop, choose by the higher NPV. When money is limited, rank by profitability. The cost belongs at time zero and is not discounted. Students often use accounting profit instead of cash flow, or forget the cost, or discount at the wrong rate. The discount rate should reflect the risk of the project.

CLOSING THE CHAPTER**SUMMARY**

Capital budgeting judges large, long-lived spending. Its steps run from proposals to review. Only incremental cash flows count. Payback is simple but ignores timing and later cash; NPV counts both and measures added wealth.

THINK IT THROUGH

A project costs Tk. 20,000 and returns Tk. 8,000 a year for 3 years. At 10%, find its payback and its NPV. Would you accept it?

WHERE PEOPLE GO WRONG

Students treat sunk costs as relevant and count depreciation as cash. Others prefer the quicker payback although the other project adds more wealth.

IN HISTORY

New York State began the Erie Canal in 1817 and opened it in 1825. It cost about seven million dollars, a vast sum for the state, and was paid for by bor-

rowing. Tolls repaid the debt early, and the canal lowered the cost of moving goods between the Atlantic coast and the Great Lakes.

REVIEW QUESTIONS

1. What is capital budgeting, and why is it necessary?
2. List the steps of capital budgeting.
3. Solve the Think it through problem.

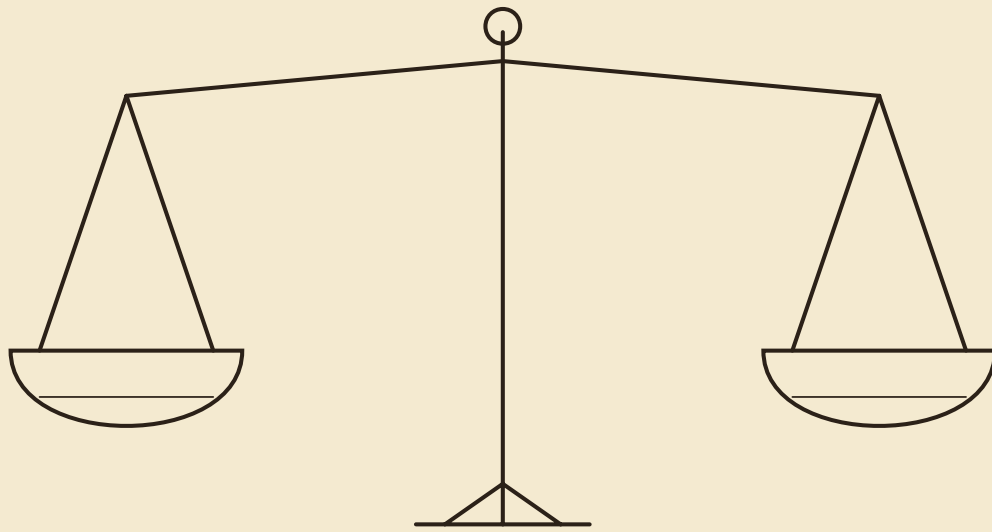


Plate IV. [An engraving will be placed here.]

CHAPTER IV

Risk and Return

Two roads reach the same place; one is safer.



This chapter explains return and risk, the kinds of risk a firm faces, how to measure risk with expected value, standard deviation and the coefficient of variation, and how risk sets the return that investors demand.

Same average, different spread.

Return is the gain; risk is doubt about the gain.

Return. The gain from an investment, shown as a percentage of the amount invested.

Risk. The chance that the actual return differs from the expected return.

Business risk comes from operations; financial risk from borrowing.

Business risk. The risk that comes from the firm's operations, such as changes in sales and costs.

Financial risk. The extra risk that comes from using borrowed money.

Risk is measured by how far returns spread from their average.

Go slowly here.

Probability distribution. A list of possible returns, each with the chance that it occurs.

Expected value. The average of possible returns, each weighted by its probability.

Standard deviation. A measure of how widely returns spread around the expected value.

Coefficient of variation. The standard deviation divided by the expected value; risk per unit of return.

A PUZZLE ABOUT TWO ROUTES

You travel to the city by two routes. Route X takes 40, 45 or 50 minutes. Route Y takes 20, 45 or 70 minutes. Both average 45 minutes. Yet if you must reach an examination on time, you choose X. Y is riskier because its results are more spread out. Finance measures this spread in the same way.

RETURN AND RISK

Return means the gain from an investment as a percentage of the amount invested. Rate of return = (selling price - buying price + cash received) / buying price. If you buy a share for Tk. 100, receive a dividend of Tk. 4 and sell for Tk. 108, the return is $(108 - 100 + 4) / 100 = 12\%$.

Risk means the chance that the actual return will differ from the one expected. Both bad and good surprises are included in the idea, though people worry about the bad.

KINDS OF RISK

Business risk means the risk that comes from the firm's operations: sales may fall, costs may rise, a rival may arrive. It exists even if the firm has no debt. **Financial risk** means the extra risk that comes from borrowing. Interest must be paid whether sales are good or poor, so debt makes the owners' return less certain. Chapter V studies this link.

Risks can also be sorted by whether they can be removed. Risk that belongs to one firm, such as a strike or a failed product, can be reduced by holding shares in many firms. Risk that affects all firms, such as a recession or a rise in interest rates, cannot. Investors are paid mainly for bearing the second kind.

MEASURING RISK

The measure uses four steps. A **probability distribution** lists the possible returns and the chance of each. The **expected value** is the average, with each return weighted by its chance. The **standard deviation** measures the spread: take each return's distance from the expected value, square it, weight it by its chance, add, and take the square root. The **coefficient of variation** divides the standard deviation by the expected value, so it shows risk for each unit of return and allows fair comparison.

WORKED EXAMPLE: INVESTMENTS X AND Y

GIVEN. three possible conditions with probabilities 0.25, 0.50, 0.25. Returns of X: 8%, 12%, 16%. Returns of Y: 4%, 12%, 20%.

FIND. expected value, standard deviation and coefficient of variation of each.

FORMULA. Expected value = sum of (probability x return). Standard deviation = square root of sum of [probability x (return - expected value)²]. CV = standard deviation / expected value.

STEPS. X: expected value = $0.25 \times 8 + 0.50 \times 12 + 0.25 \times 16 = 2 + 6 + 4 = 12\%$. Squared distances: $(8-12)^2 = 16$, 0, 16. Weighted: $0.25 \times 16 + 0 + 0.25 \times 16 = 8$. Standard deviation = square root of 8 = 2.83%. Y: expected value = $1 + 6 + 5 = 12\%$. Squared distances: 64, 0, 64. Weighted sum = 32. Standard deviation = 5.66%.

ANSWER. Both average 12%. X has standard deviation 2.83% and CV 0.24. Y has standard deviation 5.66% and CV 0.47. Y is twice as risky for the same return.

CHECK. Y's returns lie twice as far from the average as X's, and its standard deviation is twice as large.

You should now be able to say: *standard deviation tells me how widely returns spread; a larger value means more risk.*

Exceptions. Standard deviation compares well only when expected values are equal; when they differ, use the coefficient of variation. Standard deviation treats gains and losses alike. Past results do not guarantee future ones.

RISK AND THE RETURN INVESTORS DEMAND

More risk must be paid for with more return.

Risk premium. The extra return investors require for taking risk instead of a safe investment.

Required return. The minimum return an investor needs to accept an investment.

Investors dislike risk, so they require extra reward for bearing it. The **risk premium** is that extra return. The **required return** is the minimum return needed to accept an investment: $\text{required return} = \text{risk-free return} + \text{risk premium}$. The risk-free return is what a safe government security pays.

The Capital Asset Pricing Model (CAPM) sets the premium by the risk that cannot be removed. $\text{Required return} = \text{risk-free return} + \beta \times (\text{market return} - \text{risk-free return})$. Beta shows how strongly an investment moves with the whole market. With a risk-free return of 5%, a market return of 11% and a beta of 1.5, the required return is $5 + 1.5 \times (11 - 5) = 14\%$. A beta above 1 means more risk than the market as a whole.

MARKETS AND INFORMATION

Prices in an efficient market already reflect information.

Efficient market. A market in which prices already reflect the available information.

An **efficient market** is one in which prices already reflect the available information. Three forms are named. In the weak form, prices reflect all past prices, so studying old price charts brings no gain. In the semi-strong form, prices also reflect all public information, such as published accounts. In the strong form, prices reflect all information, even private. Most evidence supports the weak and, to a large degree, the semi-strong form.

CLOSING THE CHAPTER

SUMMARY

Return is the gain on an investment; risk is the chance that the gain differs from what is expected. Business and financial risk are the two main kinds in a firm. Expected value, standard deviation and the coefficient of variation measure risk. Investors demand a premium for risk.

THINK IT THROUGH

Asset M has expected return 10% and standard deviation 4%. Asset N has expected return 20% and standard deviation 6%. Which is less risky for each unit of return?

WHERE PEOPLE GO WRONG

Students forget to take the square root, or compare standard deviations when expected values differ. Others think a high return is a sign of low risk.

IN HISTORY

In 1688, according to the records, Edward Lloyd's coffee house in London was already a meeting place for shipowners, captains and merchants. Insurers gathered there to share the risks of sea voyages, and from this habit grew the market called Lloyd's.

REVIEW QUESTIONS

1. What is risk?
2. Distinguish financial risk from business risk.
3. Solve the Think it through problem.
4. A share costs Tk. 50, pays Tk. 2 and is sold for Tk. 54. Find the return.

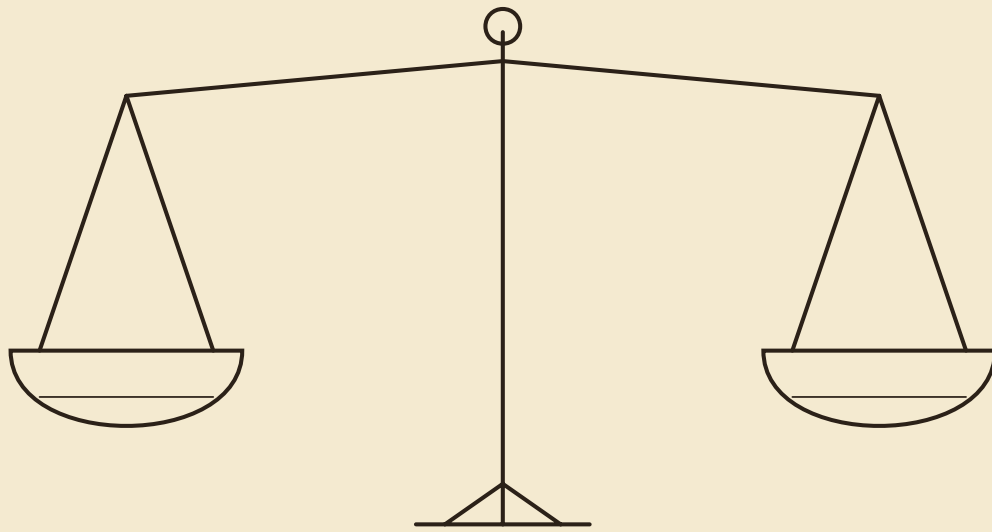


Plate V. [An engraving will be placed here.]

CHAPTER V

Leverage

A small push on one end lifts a large weight on the other.



This chapter explains operating, financial and total leverage, shows how to measure each one, and discusses why leverage can magnify both profit and trouble.

The same lever that lifts also drops.

Leverage means fixed costs magnify results.

Leverage. The use of fixed costs to magnify the effect of a change in sales on profit.

Fixed cost. A cost that stays the same whatever the level of sales.

Operating leverage. The effect of fixed operating costs on how profit before interest and tax changes with sales.

Financial leverage. The effect of fixed financing costs, such as interest, on how profit after interest responds to EBIT.

Total leverage. The combined effect of operating and financial leverage on profit after interest as sales change.

The degree of leverage is a multiplier.

Degree of operating leverage. The percentage change in EBIT caused by a one per cent change in sales.

Degree of financial leverage. The percentage change in profit after interest caused by a one per cent change in EBIT.

Degree of total leverage. The percentage change in profit after interest caused by a one per cent change in sales.

Break-even is where profit before interest and tax is zero.

Leverage raises both reward and risk.

A PUZZLE ABOUT A LEVER

A shopkeeper rents a shop for a fixed Tk. 40,000 a month and pays little for goods. Another buys goods at a high price but pays a low rent. In a month when sales rise by 10%, which shopkeeper's profit rises more? The first, since the rent does not rise with sales. In a month when sales fall by 10%, the same shopkeeper's profit falls more. A fixed cost works like a lever: it magnifies the effect of sales on profit, upward and downward.

WHAT LEVERAGE IS

Leverage means the use of fixed costs, which are costs that do not change when sales change, to magnify the effect of a change in one quantity on another. A **fixed cost** stays the same whatever the level of sales, as rent does. A variable cost rises and falls with sales, as flour does. The word comes from the lever of Archimedes.

Two kinds of fixed cost give two kinds of leverage. **Operating leverage** comes from fixed operating costs, such as rent and machines, and shows how profit before interest and tax (EBIT) responds to sales. **Financial leverage** comes from fixed financing costs, mainly interest, and shows how profit after interest (EBT) responds to EBIT. **Total leverage** combines both.

MEASURING LEVERAGE

The **degree of operating leverage** (DOL) is the percentage change in EBIT for each 1% change in sales. At a given sales level, $DOL = \text{contribution} / \text{EBIT}$, where $\text{contribution} = \text{sales} - \text{variable costs}$. The **degree of financial leverage** (DFL) = EBIT / EBT . The **degree of total leverage** (DTL) = $\text{contribution} / \text{EBT}$, which equals $DOL \times DFL$.

WORKED EXAMPLE: LEVERAGE OF A SMALL FIRM

GIVEN. price Tk. 50 per unit; variable cost Tk. 30 per unit; fixed operating cost Tk. 40,000; interest Tk. 10,000; sales 4,000 units.

FIND. DOL, DFL and DTL, and the effect of a 10% rise in sales.

FORMULA. $DOL = \text{contribution} / \text{EBIT}$; $DFL = \text{EBIT} / \text{EBT}$; $DTL = DOL \times DFL$.

STEPS. Contribution per unit = $50 - 30 = 20$. Contribution = $4,000 \times 20 = 80,000$. EBIT = $80,000 - 40,000 = 40,000$. EBT = $40,000 - 10,000 = 30,000$. $DOL = 80,000 / 40,000 = 2.00$. $DFL = 40,000 / 30,000 = 1.33$. $DTL = 80,000 / 30,000 = 2.67$, which equals 2.00×1.33 .

ANSWER. $DOL = 2.00$, $DFL = 1.33$, $DTL = 2.67$. A 10% rise in sales should raise EBT by about 26.7%.

CHECK. At 4,400 units, contribution is 88,000, EBIT is 48,000 (up 20%), EBT is 38,000 (up 26.7%). It matches.

The break-even quantity is fixed operating cost divided by contribution per unit: $40,000 / 20 = 2,000$ units. Below it, the firm makes a loss before interest.

WHY LEVERAGE MATTERS

Leverage raises the reward of good times and the damage of bad times. A firm with high operating leverage, such as one with a costly factory, is hurt badly when sales fall. A firm with high financial leverage must pay interest even in a poor year, so it may fail to pay and lose its assets. From the owner's side, debt

can raise the return on the owner's money when business is good. From the lender's side, high debt means a higher chance of not being repaid. From the manager's side, the task is to choose a level that gains from leverage without inviting failure.

Exceptions. DOL is different at each sales level, and it cannot be found at break-even, where EBIT is zero. Leverage figures assume that price, variable cost and fixed cost stay unchanged.

CLOSING THE CHAPTER

SUMMARY

Fixed costs act as a lever. Operating leverage comes from fixed operating costs; financial leverage from fixed financing costs; total leverage combines them as $DTL = DOL \times DFL$. Leverage magnifies gains and losses and therefore adds to risk.

THINK IT THROUGH

A firm sells 2,000 units at Tk. 60, with variable cost Tk. 40 a unit, fixed operating cost Tk. 20,000 and interest Tk. 5,000. Find DOL, DFL and DTL.

WHERE PEOPLE GO WRONG

Students use sales instead of contribution in DOL. Others forget that DTL is the product, not the sum, of DOL and DFL.

IN HISTORY

The Greek writer Pappus, who lived in the fourth century, reported that Archimedes had claimed he could move the Earth if given a place to stand. Whatever the truth of the saying, it made the lever a lasting picture of small force producing large effect, and financiers later borrowed the word.

REVIEW QUESTIONS

1. Define operating, financial and total leverage.
2. Why is leverage significant for a firm?
3. Solve the Think it through problem.

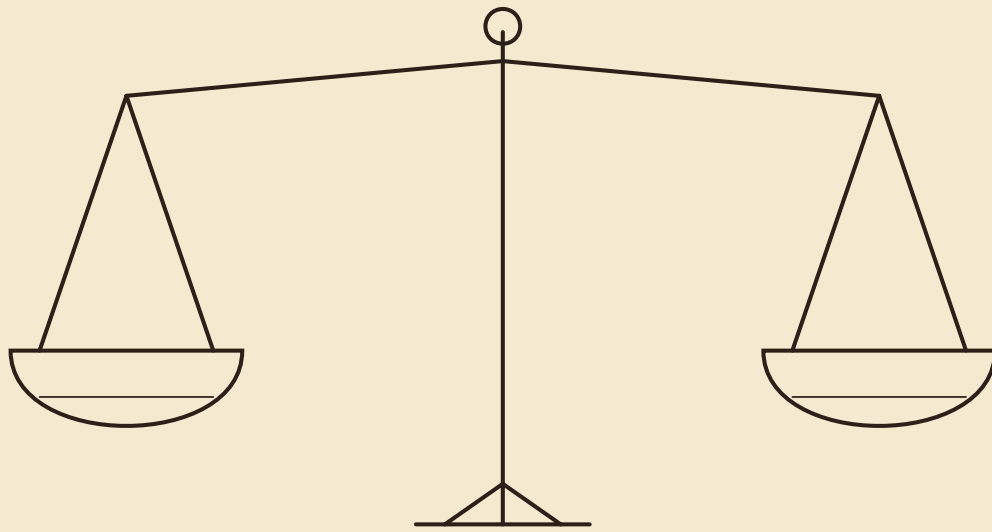


Plate VI. [An engraving will be placed here.]

CHAPTER VI

Valuing Bonds and Shares

What is a promise of future money worth today?



This chapter explains how securities are valued, how a bond is priced from its features, what its status and yield mean, how a bond differs from a share, and how a share is valued.

A paper that promises money has a price.

Value is the present value of expected cash at the required return.

Security. A paper or record that gives its holder a claim on money from a borrower or a company.

Intrinsic value. What a security is worth, found from the present value of the cash it is expected to pay.

A bond is a loan split into small, tradable pieces.

Bond. A long-term loan to a firm or government, with fixed payments, that can be bought and sold.

Face value. The sum printed on the bond and repaid at maturity; also called par value.

Coupon rate. The yearly interest rate on the face value, set when the bond is issued.

Maturity. The date on which the bond is repaid.

Required rate of return. The yield investors need from a bond of this risk, set by the market.

The bond sells at a discount, par or premium.

Bond status. Whether a bond sells below, at or above its face value.

Yield to maturity. The yearly return an investor earns by buying a bond at its price and holding it to maturity.

A PUZZLE ABOUT A PROMISE

A friend offers to sell you a paper. It promises Tk. 100 each year for three years and Tk. 1,000 at the end. How much should you pay? Not Tk. 1,300, the total promised, since later money is worth less today. The right price is the present value of everything the paper promises. This chapter applies Chapter II to real securities.

THE IDEA OF VALUE

A **security** means a paper or record that gives its holder a claim on money. **Intrinsic value** means what a security is worth, found from the present value of the cash it is expected to pay, discounted at the return investors require. If the market price is below this value, the security looks cheap; if above, it looks costly.

BONDS

A **bond** means a long-term loan to a firm or government, divided into pieces that can be traded. Its main features are these. The **face value** is the sum repaid at the end, often Tk. 1,000. The **coupon rate** is the yearly interest as a percentage of face value. The **maturity** is the date of repayment. Other features include the claim on assets if the issuer fails, and whether it can be redeemed early.

The price of a bond follows the rule of Chapter II. Price = present value of the interest payments (an annuity) + present value of the face value (a single sum), both at the required return. The **required rate of return** is set by the market for bonds of that risk.

WORKED EXAMPLE: VALUE OF A THREE-YEAR BOND

GIVEN. face value Tk. 1,000; coupon rate 10% paid yearly; 3 years; required return 12%.

FIND. the bond's value.

FORMULA. Value = interest x annuity factor + face value x single-sum factor.

STEPS. Interest = $1,000 \times 10\% = \text{Tk. } 100$. Annuity factor at 12% for 3 years = 2.40183. Single-sum factor = 0.71178. Value = $100 \times 2.40183 + 1,000 \times 0.71178 = 240.18 + 711.78$.

ANSWER. The value is Tk. 951.96.

CHECK. The coupon (10%) is below the required return (12%), so the bond must be worth less than its face value of 1,000. It is.

BOND STATUS AND YIELD

The **bond status** tells whether the bond sells at a discount (below face value), at par (equal) or at a premium (above). Compare the coupon rate with the required return. If the coupon is lower, the bond sells at a discount, because buyers must be compensated. If equal, at par. If higher, at a premium. When market rates rise, bond prices fall, and when they fall, prices rise.

The **yield to maturity** (YTM) is the yearly return earned by buying at the market price and holding to maturity. It is the discount rate that makes the present value of all payments equal to the price. In the example, a price of Tk. 951.96 gives a YTM of 12%. At a required return of 8% the same bond is worth Tk. 1051.54, a premium. At 10% it is worth Tk. 1,000, which is par.

WORKED EXAMPLE: A BOND PAYING INTEREST TWICE A YEAR

GIVEN. face value Tk. 1,000; coupon 6.5% a year paid half-yearly; 2 years; required return 8% a year.

FIND. the value.

FORMULA. Use half-yearly periods: halve the coupon and the rate, and double the years.

STEPS. Interest each half-year = $1,000 \times 6.5\% / 2 = \text{Tk. } 32.50$. Periods = 4. Rate per period = 4%. Annuity factor = 3.62990; single-sum factor = 0.85480. Value = $32.50 \times 3.62990 + 1,000 \times 0.85480 = 117.97 + 854.80$.

ANSWER. The value is Tk. 972.78.

CHECK. The coupon (6.5%) is below the required return (8%), so the value is below 1,000. It is.

SHARES

A share is a piece of ownership, with no fixed repayment.

Share. A unit of ownership in a company; also called common stock.

A **share** means a unit of ownership in a company, also called common stock. The holder receives dividends if the firm pays them and gains if the share price rises. Unlike a bond, a share has no maturity and no promised payment.

BOND AND SHARE COMPARED

Feature	Bond	Share
Holder is	A lender	An owner
Payment	Fixed interest	Dividend, not promised
Repayment	Face value at maturity	None; sell to another buyer
Voting rights	Usually none	Usually yes
Claim if firm fails	Paid before owners	Paid last
Risk to holder	Lower	Higher

Read each row across: it states how a bond and a share differ on one point. Because a share is riskier, its required return is usually higher.

The value of a share is again a present value. If the next dividend is D , the required return is r , and dividends grow forever at a steady rate g , then value = $D / (r - g)$. With $D = \text{Tk. } 5$, $r = 12\%$ and $g = 4\%$, value = $5 / (0.12 - 0.04) = \text{Tk. } 62.50$. If the dividend does not grow, value = D / r . The formula works only when r is greater than g , and it is only as good as the forecasts that go into it.

CLOSING THE CHAPTER**SUMMARY**

A security is worth the present value of the cash it is expected to pay, at the return investors require. A bond is priced from its coupon, face value, maturity and required return. Its status depends on coupon against required return. A share has no fixed payment and is valued from its expected dividends.

THINK IT THROUGH

A bond has face value Tk. 1,000, coupon 8% paid yearly, 3 years to maturity. Find its value when the required return is 10%, and state its status.

WHERE PEOPLE GO WRONG

Students use the total of payments as the price, or use the annual rate for half-yearly interest. Others think a rise in market rates raises bond prices.

IN HISTORY

In Holland in 1636 and 1637, contemporaries reported that prices for tulip bulbs rose sharply and then collapsed. Historians still debate how large and how harmful the episode was, but it remains a warning about prices that float far above any present value of expected cash.

REVIEW QUESTIONS

1. What is a bond? Name its key features.
2. How does the required return decide bond status?
3. Solve the Think it through problem.
4. How does a bond differ from a share?

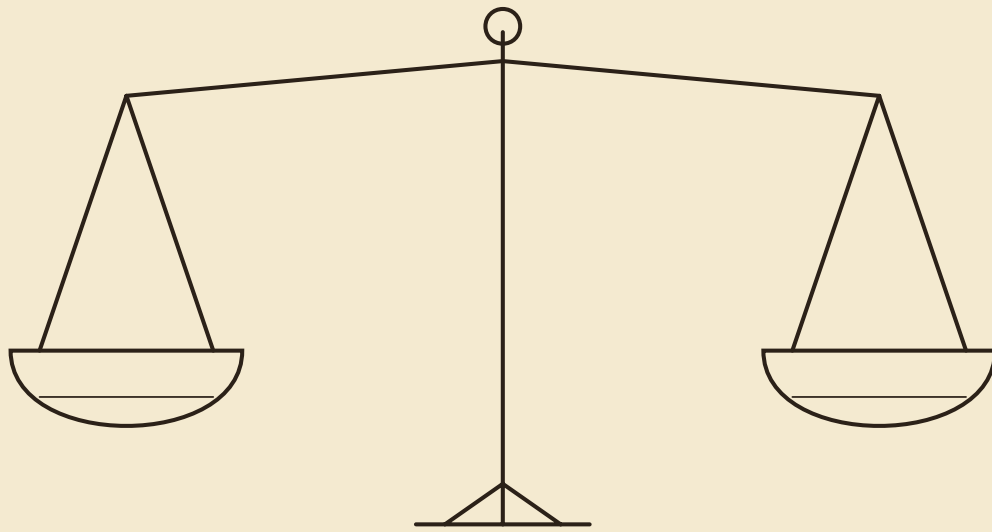


Plate VII. [An engraving will be placed here.]

CHAPTER VII

Short-Term Money

A firm can fail with a full order book and an empty till.



This chapter explains working capital, the length of financing, the matching principle, the cost of trade credit, the delay called float, and how to estimate the working capital a firm needs.

Orders do not pay wages; cash does.

Working capital is the money tied up in daily operations.

Working capital. The money a firm needs for daily operations, held in cash, stock and money owed by customers.

Liquidity. How quickly and easily an asset can be turned into cash without loss.

Financing is named by how long the money is borrowed.

Trade credit. Goods bought now and paid for later, which is a free or cheap loan from the supplier.

Match the life of the money to the life of the asset.

Matching principle. Financing an asset for about as long as the asset will be used.

A discount for early payment has a hidden interest rate.

Cash discount. A reduction in price offered for payment within a short time.

A PUZZLE ABOUT A FULL ORDER BOOK

The Bakery receives a large order from a hotel. To bake it, the Bakery must buy flour on Monday and pay wages on Friday. The hotel will pay in forty days. For forty days the Bakery spends money and receives none. If it cannot find that money, the best order of the year turns into a crisis. Short-term money is about surviving this gap.

WORKING CAPITAL

Working capital means the money a firm needs for daily operations. It is held in current assets: cash, stock of goods and the money customers owe. **Liquidity** means how quickly an asset can be turned into cash. Too little working capital risks failure to pay bills. Too much leaves money idle that could earn a return elsewhere.

SHORT, MEDIUM AND LONG FINANCING

Financing is grouped by maturity. Short-term financing is repaid within one year. Medium-term, or intermediate, financing runs from one year to about five years. Long-term financing runs for more than five years. The limits vary between books and countries, so the grouping is a guide. The next chapter studies long-term sources.

Common sources of short-term finance are these. **Trade credit** means goods bought now and paid for later. Bank overdrafts and short bank loans give quick money at a stated interest. Accrued expenses, such as wages owed but not yet paid, are a form of free credit. Firms may also borrow against money their customers owe.

THE MATCHING PRINCIPLE

The **matching principle** says to finance an asset for about as long as it will be used. A delivery van used for five years should be paid for with money repaid over about five years. Seasonal stock should be financed with short loans. If a firm finances a long-lived machine with an overdraft, the lender may demand repayment before the machine has earned the money. If it finances stock with long-term debt, it pays interest on money it does not need all year.

THE COST OF TRADE CREDIT

A supplier may say "2/10 net 30". It means: take 2% off if you pay within 10 days; otherwise pay the full sum by day 30. The **cash discount** is a reward for paying early. Not taking it is like borrowing from the supplier at a cost. Cost = $[\text{discount} / (100 - \text{discount})] \times [360 / (\text{final day} - \text{discount day})]$.

WORKED EXAMPLE: THE COST OF GIVING UP A DISCOUNT

GIVEN. terms 2/10 net 30. A bank loan costs 14% a year.

FIND. the yearly cost of trade credit, and whether to take the discount.

FORMULA. Cost = $[d / (100 - d)] \times [360 / (N - D)]$, where d is the discount per cent, D the discount day and N the final day.

STEPS. $d / (100 - d) = 2 / 98 = 0.0204$. $N - D = 30 - 10 = 20$ days, so $360 / 20 = 18$. Cost = $0.0204 \times 18 = 36.73\%$.

ANSWER. Not taking the discount costs about 36.7% a year, more than the bank's 14%. Take the discount, and borrow from the bank if cash is needed.

CHECK. For 20 days the cost is $2 / 98 = 2.04\%$; there are 18 such periods in a year, and 2.04×18 is about 36.7.

With terms 1/10 net 40 the cost is $(1 / 99) \times (360 / 30) = 12.12\%$, below 14%, so the firm should pay on day 40. Exceptions: the formula uses simple interest; with compounding the true cost is higher. Paying late costs less in the short run but may harm the firm's name with suppliers.

FLOAT AND THE CASH CYCLE

Float is the delay between paying and the money moving.

Float. The time between a payment being sent and the money becoming usable by the receiver.

Cash conversion cycle. The days between paying for materials and receiving cash from customers.

Float means the time between sending a payment and the receiver having usable funds. It has three parts: mail time, while the cheque travels; processing time, while the receiver records and deposits it; and clearing time, while the banks move the money. A firm wants short float on money it receives and, within honest limits, longer float on money it pays.

The **cash conversion cycle** shows how long money is tied up: inventory days + receivables days - payables days. If the Bakery holds stock for 40 days, waits 30 days for customers and pays suppliers after 20 days, the cycle is $40 + 30 - 20 = 50$ days.

WORKED EXAMPLE: ESTIMATING WORKING CAPITAL

GIVEN. yearly cost of sales Tk. 36,00,000; cash conversion cycle 50 days; 360 days in a year.

FIND. the working capital needed.

FORMULA. Working capital = daily cost $\times$ length of cycle in days.

STEPS. Daily cost = $36,00,000 / 360 = \text{Tk. } 10,000$. Working capital = $10,000 \times 50 = \text{Tk. } 5,00,000$.

ANSWER. The Bakery needs about Tk. 5,00,000 to run its daily operations.

CHECK. A cycle of 50 days is about $50/360$ of a year, and $50/360$ of 36,00,000 is Tk. 5,00,000.

CLOSING THE CHAPTER

SUMMARY

Working capital keeps daily operations running. Financing is short, medium or long term, and should match the life of the asset. Trade credit has a cost that can be calculated. Float delays money, and the cash conversion cycle shows how much working capital is needed.

THINK IT THROUGH

Terms are 3/10 net 45 and the bank charges 18%. Find the cost of trade credit and decide whether to take the discount.

WHERE PEOPLE GO WRONG

Students forget to divide the discount by 100 minus the discount. Others use the full credit period instead of the days after the discount period.

IN HISTORY

Francesco Datini, a merchant of Prato in Italy who died in 1410, left behind account books and a very large number of letters. They show how medieval traders used credit and bills of exchange to trade without carrying coins, an early lesson in managing the gap between paying and being paid.

REVIEW QUESTIONS

1. Define short-term, medium-term and long-term financing.
2. Find the cost of trade credit for 2/15 net 45.
3. Solve the Think it through problem.
4. What is float and what are its parts?

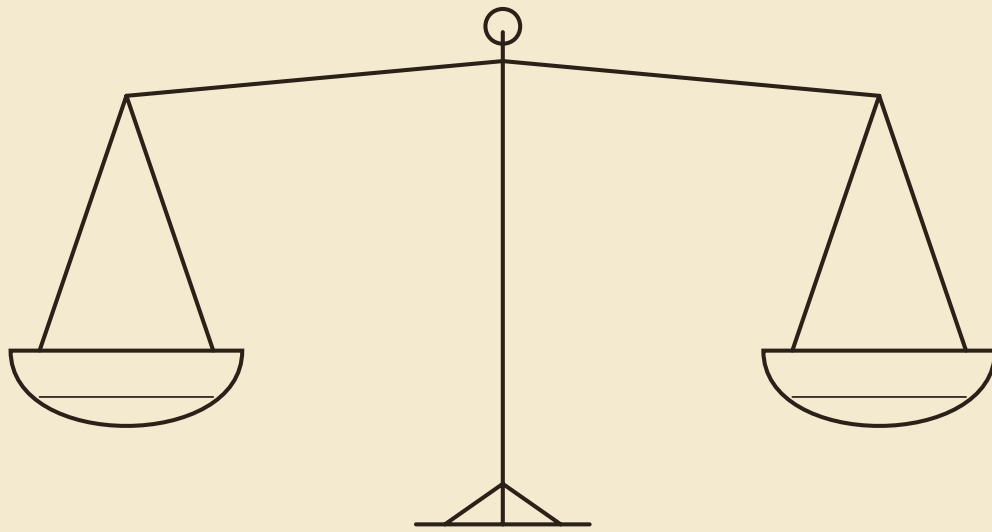


Plate VIII. [An engraving will be placed here.]

CHAPTER VIII

Long-Term Funds

A firm that plans to last must find money that lasts.



This chapter explains the long-term sources of money open to a firm: term loans, bonds, preferred shares, ordinary shares and retained profits. It shows how funds are raised in the capital market and which institutions supply them in Bangladesh.

New firms have little to offer lenders.

Six features let us compare sources.

Debt is repaid with interest on fixed dates.

Term loan. A loan from a bank or other lender for a fixed period, repaid in instalments.

Debenture. A long-term bond issued by a company, usually paying fixed interest.

Owners accept the highest risk and expect the highest return.

Preferred stock. Shares that pay a fixed dividend and rank before ordinary shares, usually without a vote.

Common stock. The ordinary shares of a company; holders vote and share in the profit left after others are paid.

Retained earnings. Profit that the firm keeps and reinvests instead of paying out as dividends.

A PUZZLE ABOUT STARTING OUT

A new firm has no profits, no record and few assets. Yet it needs a building and machines that will serve for ten years. Lenders hesitate and owners have limited savings. Where does the money come from, and what does each source ask in return? Long-term finance is the answer, and it is especially important for a new business, since a start-up's assets earn slowly and cannot be paid for from short loans.

FEATURES OF ANY LONG-TERM SOURCE

Each source can be compared on six points. Its claim on income: who is paid first. Its claim on assets if the firm fails. Its control: whether the supplier gets a vote. Its cost: the return the supplier requires. Its maturity: when repayment is due. Its risk to the firm: what happens if payment is missed. Debt is cheaper and has no vote but must be paid. Ownership money costs more but has no fixed payment.

BORROWED MONEY

A **term loan** means a loan for a fixed period, usually from one to ten years, repaid in instalments. It is common for buying machines. The lender often asks for security, which is an asset that the lender may take if the loan is not repaid. A **debenture** is a long-term bond issued by a company, which is how a company borrows from many lenders instead of one. The methods of debt thus include bank term loans, bonds and debentures.

Interest on debt is usually a cost that reduces taxable profit, so the real cost of debt is lower than the interest rate. After-tax cost of debt = interest rate $\times$ (1 - tax rate). If a firm pays 12% and the tax rate is 30% (an assumption for illustration; check the current rate), the after-tax cost is $12 \times 0.70 = 8.4\%$.

OWNERSHIP MONEY

Preferred stock means shares that pay a fixed dividend and are paid before ordinary shareholders. They are a mixture: like debt in their fixed payment, and like ownership in that a missed dividend does not bankrupt the firm.

Common stock, also called ordinary shares, means the shares of the true owners. Holders vote, receive what is left after others are paid, and bear the most risk. Their required return is therefore the highest.

Retained earnings means profit that the firm keeps instead of paying as dividends. It is the cheapest source to obtain, with no fees and no new owners, but it is not free: the owners could have used the money elsewhere, so they expect it to earn at least their required return.

THE SOURCES COMPARED

Source	Paid	Vote	Cost to firm	If payment missed
Debt	Fixed interest, first	No	Lowest after tax	Lender may take assets
Preferred stock	Fixed dividend, second	Usually no	Middle	Dividend is owed, firm survives
Common stock	What remains, last	Yes	Highest	No legal default
	Not applicable			Not applicable

Source	Paid	Vote	Cost to firm	If payment missed
Retained earnings		Existing owners	Same as common stock	

Read the table across: each row describes one source on the six features above.

RAISING MONEY IN THE CAPITAL MARKET

Large firms sell securities to many investors.

Capital market. The market where long-term funds are raised by selling shares and bonds.

Initial public offering. The first sale of a company's shares to the public.

The **capital market** is the market where long-term funds are raised and traded. A company can sell shares to the public for the first time in an **initial public offering** (IPO). It can offer new shares to its present owners in a rights issue. It can sell shares to a few large investors in a private placement. It can issue bonds or debentures. Each method needs approval from the regulator, a document describing the firm, and fees to the people who arrange the sale.

In Bangladesh, shares are traded on the Dhaka Stock Exchange and the Chittagong Stock Exchange, and the Bangladesh Securities and Exchange Commission regulates the market. Long-term finance is also supplied by commercial banks, by development finance institutions such as the Bangladesh Development Bank Limited and the Investment Corporation of Bangladesh, and by lenders for infrastructure such as IDCOL. Names and rules change, so check the current position before relying on any of this.

The **cost of capital** is the average return a firm must earn to satisfy all who supplied its long-term money. A firm should accept an investment only if it earns more than this cost. When we judge a project, we assume the project does not change the firm's business risk or financial risk; otherwise the cost of capital would change too.

Cost of capital is the average price of the firm's money.

Cost of capital. The average return the firm must earn to satisfy all those who supplied its long-term money.

CLOSING THE CHAPTER

SUMMARY

Long-term sources are debt, preferred stock, common stock and retained earnings. They differ in claim, control, cost and risk. Firms raise them in the capital market through shares, bonds and loans. The cost of capital is the average price of all of them.

THINK IT THROUGH

A firm can borrow at 15% with a tax rate of 30% or raise equity at 18%. Which is cheaper after tax, and why might the firm still not borrow without limit?

WHERE PEOPLE GO WRONG

Students say retained earnings cost nothing. Others forget that interest lowers tax and so compare before-tax debt with equity.

IN HISTORY

In 1694 the English government needed money for war with France. A group of subscribers lent it 1,200,000 pounds and in return was set up as the Bank of England. It was an early case of the public being invited to supply long-term money to a borrower.

REVIEW QUESTIONS

1. Why is long-term financing important for a new business?
2. Name the main sources of long-term money.
3. Solve the Think it through problem.

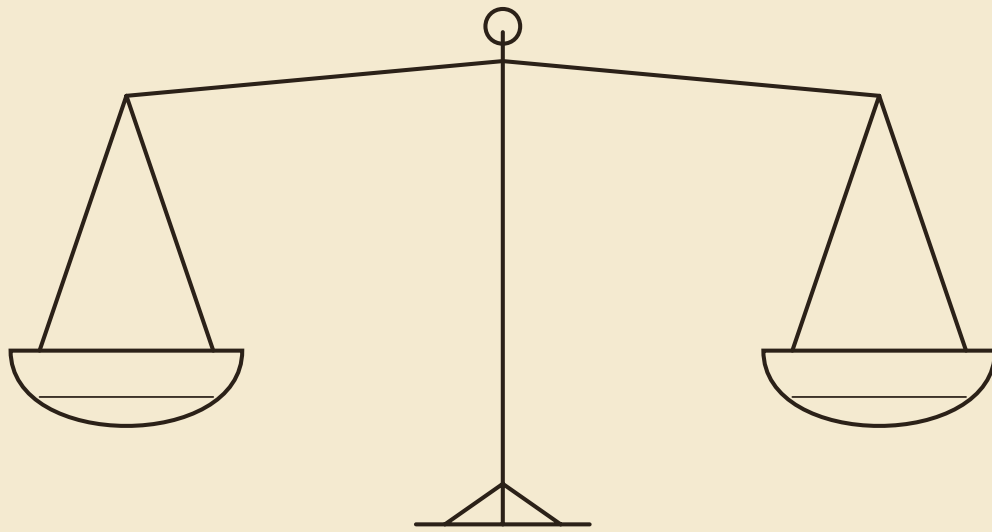


Plate IX. [An engraving will be placed here.]

CHAPTER IX

Leasing

To use a thing is not always to own it.



This chapter explains leasing, the two main kinds of lease, and the steps for deciding whether a firm should lease or buy an asset.

The Bakery needs a van but not the cost of one.

A lease is rent for the use of an asset.

Lease. A contract in which the owner of an asset lets another person use it for a fee over a set period.

Lessor. The owner of the asset who lets it to another.

Lessee. The person who uses the asset and pays the fee.

An operating lease is a short rental; a financial lease is a long commitment.

Operating lease. A lease for a short part of the asset's life, which the user can cancel, and where the owner keeps the risks of ownership.

Financial lease. A long lease that covers most of the asset's life and cannot be cancelled, so it works like a loan.

Compare the present value of the after-tax outflows.

A PUZZLE ABOUT A VAN

The Bakery needs a delivery van. Buying costs Tk. 1,00,000 at once. A dealer offers instead to let the Bakery use the van for four years for a yearly fee. Which is better? The answer depends on cost, tax and what remains at the end. To compare fairly, we must put all the cash flows on the same footing.

WHAT A LEASE IS

A **lease** means a contract in which the owner lets another person use an asset for a fee over a set period. The owner is the **lessor**. The user is the **lessee**. Leasing is a form of financing, since the lessee gets an asset without paying its full price now.

TWO KINDS OF LEASE

An **operating lease** lasts for part of the life of the asset, can usually be cancelled, and leaves the lessor with the costs of repairs and with the asset at the end. A **financial lease** covers most of the life of the asset, usually cannot be cancelled, and the lessee bears repairs and similar costs. It works like a loan used to buy the asset.

OPERATING AND FINANCIAL LEASES

Point	Operating lease	Financial lease
Length	Part of asset's life	Most of asset's life
Cancellation	Usually allowed	Usually not allowed
Repairs	Lessor pays	Lessee pays
Total payments	Less than the asset's cost	About the cost plus interest
Best for	Short or changing needs	Long, steady needs

Read each row across: it compares the two leases on one point. The reasons to lease include less money paid at the start, flexibility, and sometimes tax gain. The drawbacks include a higher total cost and, at the end, no ownership.

THE LEASE OR BUY DECISION

The decision has four steps. Step one: find the after-tax cash outflows of leasing. Step two: find those of buying. Step three: discount both at the after-tax cost of debt, because both are in effect ways of borrowing. Step four: choose the plan with the lower present value of outflows. Lease payments reduce taxable profit; so do the interest and depreciation of a purchase. Tax rules differ between countries, so state the rules used.

WORKED EXAMPLE: LEASE OR BUY A VAN

GIVEN. price Tk. 1,00,000; 4 years. Lease payment Tk. 30,000 at each year end. Purchase by a 10% loan with instalments of Tk. 31,500 a year. Depreciation Tk. 25,000 a year. Interest: Year 1 10,000, Year 2 7,500, Year 3 5,000, Year 4 2,500. Tax rate 30% (assumed).

FIND. which plan has the lower present value of after-tax outflows.

FORMULA. After-tax lease cost = payment $\times$ (1 - tax rate). After-tax purchase cost = instalment - tax rate $\times$ (depreciation + interest). Discount rate = 10% $\times$ (1 - 0.30) = 7%.

STEPS. Lease: $30,000 \times 0.70 = 21,000$ each year. $PV = 21,000 \times 3.38721 = 71,131$.
Purchase outflows: Year 1: $31,500 - 0.30 \times 35,000 = 21,000$; Year 2: $31,500 - 0.30 \times 32,500 = 21,750$; Year 3: $31,500 - 0.30 \times 30,000 = 22,500$; Year 4: $31,500 - 0.30 \times 27,500 = 23,250$. $PV = 74,727$.

ANSWER. Leasing has the lower present value, Tk. 71,131 against Tk. 74,727 for buying, so lease the van.

CHECK. Each yearly lease outflow (21,000) is below or equal to each purchase outflow, so leasing must have the lower present value. This test is not always possible; here it is.

Exceptions: this comparison leaves out the value of the van at the end of four years, which a buyer keeps. If that value is large, buying may win. The residual value must be estimated and added. Maintenance, insurance and risk of the asset becoming old also matter.

CLOSING THE CHAPTER

SUMMARY

A lease lets the lessee use an asset owned by the lessor. Operating leases are short and cancellable; financial leases are long and work like loans. The lease-or-buy decision compares the present value of after-tax outflows at the after-tax cost of debt.

THINK IT THROUGH

Explain why a firm needing a computer for six months would prefer an operating lease to a financial lease.

WHERE PEOPLE GO WRONG

Students discount at the before-tax rate, or forget the tax saving on depreciation and interest. Others ignore the value of the asset at the end.

IN HISTORY

In 1952 a firm in San Francisco, the United States Leasing Corporation, began to lease equipment to businesses that wanted to use machines without buying them. Leasing then spread as a standard way of financing equipment.

REVIEW QUESTIONS

1. What is leasing? Distinguish operating from financial leases.
2. State the four steps of the lease-or-buy decision.
3. Why is the after-tax cost of debt used as the discount rate?

Answers to Review Questions

CHAPTER I

1. Finance is the way a business finds, uses and controls money; it asks where money comes from, where it is placed and how its use is judged.
2. Investment (what to buy), financing (where money comes from), working capital (daily money), dividend (how much profit to pay out).
3. It ignores timing, risk and the difference between profit and cash, and may harm the future; wealth maximisation counts all three.

CHAPTER II

1. A sum today is worth more than the same sum later, because money can earn, prices rise and the future is uncertain.
2. A series of equal payments at equal intervals; ordinary (end of period), due (start of period), and perpetuity (never ends).
3. $P = \text{Tk. } 12,434.26$; $Q = \text{Tk. } 10,383.17$. P is worth more today.
4. $1,00,000 \times 5.8666 = \text{Tk. } 5,86,660$.

CHAPTER III

1. Planning and judging large, long-lived spending; necessary because sums are large, benefits long, decisions hard to reverse and the future of the firm depends on them.
2. Generate proposals; estimate cash flows; evaluate; select; implement; review.
3. $\text{Payback} = 20,000 / 8,000 = 2.5$ years. $\text{NPV} = 8,000 \times 2.48685 - 20,000 = \text{Tk. } -105.18$. It is negative, so reject.

CHAPTER IV

1. The chance that the actual return differs from the expected return.
2. Business risk comes from operations and exists without debt; financial risk is the extra risk from borrowing.
3. $\text{CV of } M = 4/10 = 0.40$; $\text{CV of } N = 6/20 = 0.30$. N is less risky for each unit of return.
4. $(54 - 50 + 2) / 50 = 12\%$.

CHAPTER V

1. Operating: fixed operating costs magnify the effect of sales on EBIT. Financial: fixed financing costs magnify the effect of EBIT on EBT. Total: the two combined, from sales to EBT.
2. It magnifies both profit and loss, so it raises risk; managers must balance the gain against the danger of failing to meet fixed costs.
3. $\text{Contribution} = 2,000 \times 20 = 40,000$; $\text{EBIT} = 20,000$; $\text{EBT} = 15,000$. $\text{DOL} = 2.00$; $\text{DFL} = 1.33$; $\text{DTL} = 2.67$.

CHAPTER VI

1. A long-term loan split into tradable pieces; face value, coupon rate and maturity are its key features.
2. Coupon below required return: discount. Equal: par. Above: premium.

3. $80 \times 2.48685 + 1,000 \times 0.75131 = \text{Tk. } 950.26$; it sells at a discount.

4. A bond is a loan with fixed payments and a repayment date; a share is ownership with no promised payment and no maturity.

CHAPTER VII

1. Repaid within one year; one to about five years; more than five years.

2. $(2/98) \times (360/30) = 24.49\%$.

3. $(3/97) \times (360/35) = 31.81\%$. This exceeds 18%, so take the discount.

4. The delay between sending a payment and the money becoming usable; mail, processing and clearing time.

CHAPTER VIII

1. A new firm needs lasting assets that earn slowly; short loans would fall due before the assets repay them.

2. Term loans, bonds or debentures, preferred stock, common stock, retained earnings.

3. Debt: $15 \times 0.70 = 10.5\%$, cheaper than equity at 18%. Heavy debt raises financial risk, so lenders and owners demand more.

CHAPTER IX

1. Leasing lets one person use an asset owned by another for a fee. Operating leases are short, cancellable, and the lessor bears costs; financial leases are long, not cancellable, and work like loans.

2. After-tax outflows of lease; after-tax outflows of purchase; discount both at the after-tax cost of debt; choose the lower present value.

3. Both plans are in effect borrowing, and interest saves tax, so the after-tax cost of borrowing is the fair rate.

Glossary

Agency costs. The costs of watching agents and of the losses when they act against the owners. (*Chapter I*)

Agency problem. The risk that agents act for themselves instead of for the owners. (*Chapter I*)

Agent. A person who acts on behalf of another. (*Chapter I*)

Annuity. A series of equal payments made at equal intervals. (*Chapter II*)

Annuity due. An annuity whose payments are made at the start of each period. (*Chapter II*)

Bond. A long-term loan to a firm or government, with fixed payments, that can be bought and sold. (*Chapter VI*)

Bond status. Whether a bond sells below, at or above its face value. (*Chapter VI*)

Business risk. The risk that comes from the firm's operations, such as changes in sales and costs. (*Chapter IV*)

Capital budgeting. Planning and judging large spending whose benefits last for several years. (*Chapter III*)

Capital market. The market where long-term funds are raised by selling shares and bonds. (*Chapter VIII*)

Cash conversion cycle. The days between paying for materials and receiving cash from customers. (*Chapter VII*)

Cash discount. A reduction in price offered for payment within a short time. (*Chapter VII*)

Cash flow. Money actually received or paid by the firm. (*Chapter III*)

Coefficient of variation. The standard deviation divided by the expected value; risk per unit of return. (*Chapter IV*)

Common stock. The ordinary shares of a company; holders vote and share in the profit left after others are paid. (*Chapter VIII*)

Compound interest. Interest calculated on the original sum plus the interest already earned. (*Chapter II*)

Cost of capital. The average return the firm must earn to satisfy all those who supplied its long-term money. (*Chapter VIII*)

Coupon rate. The yearly interest rate on the face value, set when the bond is issued. (*Chapter VI*)

Debenture. A long-term bond issued by a company, usually paying fixed interest. (*Chapter VIII*)

Degree of financial leverage. The percentage change in profit after interest caused by a one per cent change in EBIT. (*Chapter V*)

Degree of operating leverage. The percentage change in EBIT caused by a one per cent change in sales. (*Chapter V*)

Degree of total leverage. The percentage change in profit after interest caused by a one per cent change in sales. (*Chapter V*)

Discount rate. The yearly rate used to reduce a future sum to its present value. (*Chapter II*)

Dividend. A share of profit paid out to the owners of a company. (*Chapter I*)

Efficient market. A market in which prices already reflect the available information. (*Chapter IV*)

Expected value. The average of possible returns, each weighted by its probability. (*Chapter IV*)

- Face value.** The sum printed on the bond and repaid at maturity; also called par value. *(Chapter VI)*
- Finance.** The way a business finds, uses and controls its money. *(Chapter I)*
- Financial lease.** A long lease that covers most of the asset's life and cannot be cancelled, so it works like a loan. *(Chapter IX)*
- Financial leverage.** The effect of fixed financing costs, such as interest, on how profit after interest responds to EBIT. *(Chapter V)*
- Financial manager.** The person who decides how a firm finds and uses its money. *(Chapter I)*
- Financial risk.** The extra risk that comes from using borrowed money. *(Chapter IV)*
- Financing decision.** A choice about how to raise money, from owners or from lenders. *(Chapter I)*
- Fixed cost.** A cost that stays the same whatever the level of sales. *(Chapter V)*
- Float.** The time between a payment being sent and the money becoming usable by the receiver. *(Chapter VII)*
- Future value.** What a sum of money today will be worth at a later date, after earning interest. *(Chapter II)*
- Incremental.** Extra; the change in cash caused by choosing the project. *(Chapter III)*
- Initial public offering.** The first sale of a company's shares to the public. *(Chapter VIII)*
- Interest.** The payment a borrower makes for the use of money, shown as a percentage of the sum. *(Chapter II)*
- Intrinsic value.** What a security is worth, found from the present value of the cash it is expected to pay. *(Chapter VI)*
- Investment decision.** A choice about which assets, such as machines or shops, the firm should buy. *(Chapter I)*
- Lease.** A contract in which the owner of an asset lets another person use it for a fee over a set period. *(Chapter IX)*
- Lessee.** The person who uses the asset and pays the fee. *(Chapter IX)*
- Lessor.** The owner of the asset who lets it to another. *(Chapter IX)*
- Leverage.** The use of fixed costs to magnify the effect of a change in sales on profit. *(Chapter V)*
- Liquidity.** How quickly and easily an asset can be turned into cash without loss. *(Chapter VII)*
- Matching principle.** Financing an asset for about as long as the asset will be used. *(Chapter VII)*
- Maturity.** The date on which the bond is repaid. *(Chapter VI)*
- Mixed stream.** A series of payments that are not all equal. *(Chapter II)*
- Net present value.** The present value of a project's cash inflows minus its initial cost. *(Chapter III)*
- Operating lease.** A lease for a short part of the asset's life, which the user can cancel, and where the owner keeps the risks of ownership. *(Chapter IX)*
- Operating leverage.** The effect of fixed operating costs on how profit before interest and tax changes with sales. *(Chapter V)*
- Ordinary annuity.** An annuity whose payments are made at the end of each period. *(Chapter II)*
- Payback period.** The time a project takes to return its initial cost in cash. *(Chapter III)*
- Preferred stock.** Shares that pay a fixed dividend and rank before ordinary shares, usually without a vote. *(Chapter VIII)*
- Present value.** What a sum due in the future is worth today. *(Chapter II)*
- Principal.** A person who asks another person to act for them. *(Chapter I)*
- Probability distribution.** A list of possible returns, each with the chance that it occurs. *(Chapter IV)*

- Profit maximisation.** Aiming for the largest possible profit. (*Chapter I*)
- Required rate of return.** The yield investors need from a bond of this risk, set by the market. (*Chapter VI*)
- Required return.** The minimum return an investor needs to accept an investment. (*Chapter IV*)
- Retained earnings.** Profit that the firm keeps and reinvests instead of paying out as dividends. (*Chapter VIII*)
- Return.** The gain from an investment, shown as a percentage of the amount invested. (*Chapter IV*)
- Risk.** The chance that the actual return differs from the expected return. (*Chapter IV*)
- Risk premium.** The extra return investors require for taking risk instead of a safe investment. (*Chapter IV*)
- Security.** A paper or record that gives its holder a claim on money from a borrower or a company. (*Chapter VI*)
- Share.** A unit of ownership in a company; also called common stock. (*Chapter VI*)
- Shareholders.** The people who own shares, and so own part of a company. (*Chapter I*)
- Standard deviation.** A measure of how widely returns spread around the expected value. (*Chapter IV*)
- Sunk cost.** Money already spent that cannot be recovered whatever is decided. (*Chapter III*)
- Term loan.** A loan from a bank or other lender for a fixed period, repaid in instalments. (*Chapter VIII*)
- Time value of money.** The idea that a sum of money is worth more now than the same sum later. (*Chapter II*)
- Total leverage.** The combined effect of operating and financial leverage on profit after interest as sales change. (*Chapter V*)
- Trade credit.** Goods bought now and paid for later, which is a free or cheap loan from the supplier. (*Chapter VII*)
- Wealth maximisation.** Aiming to make the owners' investment as valuable as possible. (*Chapter I*)
- Working capital.** The money a firm needs for daily operations, held in cash, stock and money owed by customers. (*Chapter VII*)
- Yield to maturity.** The yearly return an investor earns by buying a bond at its price and holding it to maturity. (*Chapter VI*)

Further Reading

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Gitman, L. *Principles of Managerial Finance*. Many worked examples.

Jensen, M. and Meckling, W. (1976). Theory of the firm: managerial behavior, agency costs and ownership structure. *Journal of Financial Economics*. The founding paper of agency theory.

Smith, A. (1776). *An Inquiry into the Nature and Causes of the Wealth of Nations*.

Maths You Will Use

PERCENTAGES

A percentage is a number out of 100. To find 12% of Tk. 500, multiply: $500 \times 0.12 = \text{Tk. } 60$. To turn 12% into a decimal, divide by 100.

PERCENTAGE CHANGE

Percentage change = $(\text{new value} - \text{old value}) / \text{old value} \times 100$. A rise from 40,000 to 48,000 is $8,000 / 40,000 = 0.20$, which is 20%.

RATIOS AND AVERAGES

A ratio compares two numbers by division. An average is the sum of the values divided by how many values there are.

REARRANGING AN EQUATION

Whatever you do to one side, do to the other. From $FV = PV \times 1.1$, divide both sides by 1.1 to get $PV = FV / 1.1$.

POWERS

1.1 to the power 3 means $1.1 \times 1.1 \times 1.1 = 1.331$. A negative power means division: 1.1 to the power -3 is $1 / 1.331 = 0.7513$.

ROUNDING

Keep at least four decimal places in factors, and round money only in the last step. Small differences in the last digit are normal.

READING A GRAPH

The horizontal axis shows one quantity, such as time. The vertical axis shows another, such as money. Each point shows the two values together.



PRINCIPLES TO FINANCE

BIJOY?!